an AI system promotes discriminatory race-based stereotypes is a focus of safety evaluation. However, treating social constructs such as race as *fixed attributes* in evaluation may create a distorted view of the actual differential impacts on different racial groups and intersectionalities, such as with social class (Hanna et al., 2020). Similarly, some harms are particularly difficult to trace, even in hindsight – such as the long-suspected and now-evidenced link between social media and teenage eating disorders (Bulimia Project). Where effect sizes are small and causal mechanisms poorly understood, evaluation may fail to detect risks that it seeks to measure. This may also affect the detection of potential emergent capabilities that may only become observable as an AI system reaches a certain scale. In addition, overall impacts of an AI system on complex notions such as welfare are difficult to measure because the target construct itself (welfare) is difficult to establish. Especially where effects are distributed and interact with other factors such as user vulnerabilities, it can be difficult to establish hard findings. Long-term and mixed-methods approaches, including initial qualitative work, can help reduce these limitations and shed light on potential subtle or highly indirect effects.

In sum, even with best efforts, there will always be harms of particular kinds or in particular contexts that are not evaluated. This is why evaluation must be complemented with effective governance mechanisms for evaluating remaining uncertainties prior to AI system release, with post-deployment monitoring including logging observed incidents (AI Incident Database) and with well-functioning and swift recourse mechanisms for people who experience or detect harm. It is important that AI systems are flexibly designed such that new insights can be translated into fixes, such as via system updates. Given the pre-deployment evaluation gaps, organisations deploying AI systems require adequate governance infrastructures that can respond to detected risks with mitigations, by delaying or stopping the deployment of an AI system or by suspending an already-deployed system until concerns are resolved.

Generative AI systems pose significant risks for individuals, communities, and society, and failing to detect and mitigate such risks can have serious consequences (Xiang, 2023). This is why it is critical to ensure that evaluation is prioritised and that complementary mechanisms exist to uphold AI system safety to cover the gaps that are inherent limitations to evaluation.

5.3.2. Evaluation is never value-neutral

Evaluations are inherently value expressions of those who conduct them: they always require a decision on what is valued (Bowker and Star, 2000). It is widely understood and expressed in the sociotechnical literature that AI systems are not merely mathematical constructs but sociotechnical and political entities, with inherent value systems embedded in the choices made by designers with respect to how to create and implement the model (Barocas et al., 2019; Birhane et al., 2022b; Gururangan et al., 2022; Raji et al., 2021; Sambasivan et al., 2021; Scheuerman et al., 2021; Suresh and Guttag, 2021). The same extends to evaluations of an AI model or system.

There are normative decisions throughout the evaluation construction and implementation process that cannot be avoided. No evaluation can cover every circumstance and dimension of what can be evaluated (Bergman et al., 2023; Raji et al., 2021). Thus, designing an evaluation involves choices – made either deliberately or implicitly – on what to prioritise and what to discard. First, selecting a target to evaluate requires a normative judgement on what harms are important or relevant to measure (Kalluri, 2020; Mohamed et al., 2020). Operationalising the harm further requires normative decisions on what task is most valuable for the system to perform highly on, what high performance looks like, and where or to whom it is most valuable/optimal for the benefits of the system to accrue. After this process, what remains within scope of the evaluation is what is prioritised, and these decisions inherently express value.

Furthermore, operationalising a harm construct into a metric necessarily bakes in

certain assumptions. For example, making a commitment to a test and a metric – e.g. that "social biases" can be measured via associations of gender and occupations – is a normative judgement on where harms are likely to occur and which particular aspects of a harm are relevant and tractable (Luccioni et al., 2023). These normative decisions are all the more significant, as they tend to have a sticking effect that propagates (see Practical steps to closing the multimodal evaluation gap).

Assessing whether a model meets expectations prior to or post deployment requires a normative evaluation of whether some measurement expresses performance that is "good", "bad", "safe enough", etc. (see Bakalar et al. (2021)). For such thresholds to be legitimate, they need to arise from adequate institutions or processes, such as expert groups, democratic institutions, or fair and inclusive deliberation processes that centre groups that may be affected by these AI systems. The thresholds of what constitutes "acceptable" model performance may differ across use cases, contexts, or applications.[10]

Another aspect in which evaluations express value is who and whose perspectives are represented in the evaluation – for example, the English-speaking or Western world (DeVries et al., 2019; Gururangan et al., 2022; Shankar et al., 2017); tech-savvy and educated ML designers; socioeconomically privileged users providing feedback on a system; or annotators that skew young, female, educated, and white (Ding et al., 2022). Some harms will be missed if communities that may be affected are overlooked or dimensions of harm are ignored. Calls for greater representation of community groups is widespread and often offered as a mitigation for a broad range of fairness and sociotechnical harms (e.g. Costanza-Chock (2020); DeVries et al. (2019); European Commission (2021); Jindal (2021); Lashbrook (2018); National Institute of Standards and Technology (2021b); Pasquale and Malgieri (2021); Sortition Foundation (2023); Suresh and Guttag (2021)). When a community is missing in evaluation, there is no pre-deployment evaluation of the impacts of the system on that community, which can lead to skewed decision-making that ignores potential effects on these groups (Bergman et al., 2023; Buolamwini and Gebru, 2018). While greater inclusion of groups and perspectives in the evaluation can lead to better visibility of the model performance (Buolamwini and Gebru, 2018), there are arguments for adopting this approach with care (Bergman et al., 2023), as, for example, an oversimplified interpretation of this notion can lead to objectification and exploitation (e.g. Fussell (2019)).

Normative decisions on AI systems that affect large groups of people ought to be made in legitimate and accountable ways. These decisions should be dynamic such that they can be updated over time and adapted to specific application domains or locales within some range of acceptability. As a first step, identifying normative choices and documenting them can provide the basis for accountability. Second, the acknowledgement that normative choices are not merely pragmatic or technical provides a foundation for shared responsibility and public engagement on certain decisions pertaining to AI safety (e.g. what should be evaluated). Ideally, significant normative decisions should be made via deliberate processes that employ inclusive, participatory techniques (Birhane et al., 2022a). In addition to providing legitimacy, such processes may result in more robust evaluations that better track the most significant risks of downstream harms from AI systems. Documentation of how such decisions are made serves to provide further transparency and provides insights into potential limitations of evaluations (Raji et al., 2021).

[10]For example, generative models may be used in an application that provides access to news content or in a creative collaborator tool. Factually incorrect outputs may be of greater concern in the former than in the latter, depending on contexts such as the user expectations and de facto uses of the model (i.e. whether people assume and rely on the model as a source of truth). While the same evaluation for factually incorrect outputs can be run on these different use cases, different thresholds of acceptable levels of model performance may be applied. This requires performing general evaluations as early as possible and transparently disclosing results such that downstream users or product developers can make informed decisions on whether the model is fit for their intended purposes.

5.4. Steps forward

For evaluation to be impactful, four conditions must be met. First, evaluations on relevant risks of harm must exist. Second, these evaluations must be conducted regularly. Third, the conduct of these evaluations must have teeth, i.e. have meaningful consequences. Fourth, the conduct of these evaluations must become increasingly standardised and independent to ensure valid evaluations over time.

5.4.1. Evaluations must be developed where they do not yet exist

A review of the current state of safety evaluations suggests a pressing gap exists in the collective safety evaluation toolkit. This gap will likely be further exacerbated as increasingly capable multimodal models are released more broadly at pace. In addition, evaluations are often conducted in an ad hoc manner and too late to anticipate preventable harm. This assessment calls for a strong focus, concerted action, and shared priority to develop safety evaluation.

5.4.2. Evaluations must be done as a matter of course

Evaluations must be conducted during the process of AI development, to bake in ethical and social considerations from the inception of an AI system rather than imperfectly patching them on as an afterthought. In particular, evaluations should be conducted from the moment of planning a new AI system. In the context of training large generative AI systems, early indicators of risks of harm can already be obtained from analysing technical components, such as training data, or a small, raw, pre-trained model. Adequate safety testing evaluates these components to influence responsible decision-making, such as by indicating whether a training dataset is appropriate for use or whether a model training course should be continued. Early testing allows decision-makers to verifiably consider the best information available at the time of making consequential decisions. Early evaluation is also a basic tenet of safety engineering, as it supports risk mitigation and the "cost of fixing" a hazard tends to increase exponentially with time in a system's life cycle (Leveson, 2012).

Evaluation must also be a continuous practice. Evaluation is subject to Collingridge's Dilemma, which asserts that early-stage evaluation creates higher potential to influence the direction of a technology but later-stage evaluation provides more accurate and comprehensive information (Collingridge, 1982). In other words, early evaluation is less accurate but has the capacity to shape AI development more deeply. This dilemma may be somewhat modulated in the case of software systems that can, at some cost, be updated over time and post deployment. Yet to ensure that evaluation can both shape early decision-making and provide an accurate picture of risks of harm, it must be conducted at multiple time points throughout the AI system life cycle, including by monitoring effects post deployment.

The frequency of evaluations is subject to a further tension described by Goodhart's Law, which asserts that a measure that becomes a target ceases to be a good measure. Evaluations that are run frequently, e.g. as a method of performance tracking for AI *development*, become de facto targets over time. AI designers aim to improve AI system performance on these particular metrics. To provide valid *assurances* on AI system safety, evaluations cannot be used as de facto targets. To achieve this, it is important that assurance evaluations are not shared with or reverse-engineerable by AI system developers. This, in turn, requires meaningful separation between development and assurance evaluations, and between the actors who develop and those who evaluate AI systems. These separations are overall best practice in evaluation but especially important on assurance evaluations.

Finally, AI evaluations in practice face a "realism trade-off" between the pragmatic costs of conducting an evaluation on the one hand and the accuracy and validity of results on the other (Liao and Xiao, 2023). This means that evaluations sit on a spectrum between yielding high accuracy – such as longitudinal, highly localised ethnographic studies – and being automated and highly generalisable – such as automated tests and benchmarks. In other